

```
CREATE DATABASE BankDB;
```

```
USE BankDB;
```

```
=====
```

## 2. CREATE CUSTOMER TABLE

```
=====
```

```
CREATE TABLE Customer (  
    Customer_ID INT PRIMARY KEY,  
    Customer_Name VARCHAR(100) NOT NULL,  
    Phone VARCHAR(15),  
    Email VARCHAR(100),  
    City VARCHAR(50)  
);
```

```
=====
```

## 3. CREATE ACCOUNT TABLE

```
=====
```

```
CREATE TABLE Account (  
    Account_No INT PRIMARY KEY,  
    Customer_ID INT,  
    Account_Type VARCHAR(20),  
    Balance DECIMAL(12,2),  
    Branch VARCHAR(50),  
  
    FOREIGN KEY (Customer_ID)
```

```
REFERENCES Customer(Customer_ID)
);
```

```
=====
4. CREATE BANK TRANSACTION TABLE
=====
```

```
CREATE TABLE Bank_Transaction (
    Transaction_ID INT PRIMARY KEY AUTO_INCREMENT,
    Account_No INT,
    Transaction_Type VARCHAR(20),
    Amount DECIMAL(12,2),
    Transaction_Date DATETIME DEFAULT CURRENT_TIMESTAMP,

    FOREIGN KEY (Account_No)
    REFERENCES Account(Account_No)
);
```

```
=====
5. CREATE LOAN TABLE
=====
```

```
CREATE TABLE Loan (
    Loan_ID INT PRIMARY KEY,
    Customer_ID INT,
    Loan_Type VARCHAR(30),
    Loan_Amount DECIMAL(12,2),
    Interest_Rate DECIMAL(5,2),
```

```
FOREIGN KEY (Customer_ID)
REFERENCES Customer(Customer_ID)
);
```

```
=====
6. INSERT CUSTOMER DATA
=====
```

```
INSERT INTO Customer
(Customer_ID, Customer_Name, Phone, Email, City)
VALUES
(101, 'Ravi Kumar', '9876543210',
'ravi@gmail.com', 'Hyderabad'),

(102, 'Priya Sharma', '9876543211',
'priya@gmail.com', 'Vijayawada'),

(103, 'Arjun Reddy', '9876543212',
'arjun@gmail.com', 'Bangalore'),

(104, 'Sneha Rao', '9876543213',
'sneha@gmail.com', 'Chennai'),

(105, 'Kiran Kumar', '9876543214',
'kiran@gmail.com', 'Hyderabad'),

(106, 'Anil Kumar', '9876543215',
'anil@gmail.com', 'Delhi'),

(107, 'Meena Reddy', '9876543216',
```

'meena@gmail.com', 'Mumbai'),

(108, 'Rahul Sharma', '9876543217',

'rahul@gmail.com', 'Pune'),

(109, 'Lakshmi Devi', '9876543218',

'lakshmi@gmail.com', 'Hyderabad'),

(110, 'Suresh Babu', '9876543219',

'suresh@gmail.com', 'Vijayawada');

The screenshot shows a SQL IDE interface. The top toolbar includes icons for file operations, execution, and settings. The SQL editor contains the following code:

```
46 (108, 'Rahul Sharma', '9876543217', 'rahul@gmail.com', 'Pune'),
47 (109, 'Lakshmi Devi', '9876543218', 'lakshmi@gmail.com', 'Hyderabad'),
48 (110, 'Suresh Babu', '9876543219', 'suresh@gmail.com', 'Vijayawada');
49 • select * from Customer
50 ❌ INSERT INTO Account
```

Below the editor is the 'Result Grid' showing a table with 5 columns: Customer\_ID, Customer\_Name, Phone, Email, and City. The data is as follows:

| Customer_ID | Customer_Name | Phone      | Email             | City       |
|-------------|---------------|------------|-------------------|------------|
| 106         | Anil Kumar    | 9876543215 | anil@gmail.com    | Delhi      |
| 103         | Arjun Reddy   | 9876543212 | arjun@gmail.com   | Bangalore  |
| 105         | Kiran Kumar   | 9876543214 | kiran@gmail.com   | Hyderabad  |
| 109         | Lakshmi Devi  | 9876543218 | lakshmi@gmail.com | Hyderabad  |
| 107         | Meena Reddy   | 9876543216 | meena@gmail.com   | Mumbai     |
| 102         | Priya Sharma  | 9876543211 | priya@gmail.com   | Vijayawada |

Below the result grid is the 'Output' window, which shows the execution log:

| #   | Time     | Action                                                                          | Message                                                            |
|-----|----------|---------------------------------------------------------------------------------|--------------------------------------------------------------------|
| ❌ 1 | 15:36:24 | INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email, City) VALUES... | Error Code: 1062. Duplicate entry '101' for key 'customer.PRIMARY' |
| ✅ 2 | 15:37:47 | select * from Customer LIMIT 0, 1000                                            | 10 row(s) returned                                                 |

=====

## 7. INSERT ACCOUNT DATA

=====

INSERT INTO Account

(Account\_No, Customer\_ID, Account\_Type, Balance, Branch)

VALUES

(10001, 101, 'Savings', 50000, 'Hyderabad'),

(10002, 102, 'Savings', 75000, 'Vijayawada'),

(10003, 103, 'Current', 120000, 'Bangalore'),

(10004, 104, 'Savings', 45000, 'Chennai'),

(10005, 105, 'Current', 90000, 'Hyderabad'),

(10006, 106, 'Savings', 65000, 'Delhi'),

(10007, 107, 'Current', 150000, 'Mumbai'),

(10008, 108, 'Savings', 35000, 'Pune'),

(10009, 109, 'Savings', 85000, 'Hyderabad'),

(10010, 110, 'Current', 110000, 'Vijayawada');

```

59      (10007, 107, 'Current', 150000, 'Mumbai'),
60      (10008, 108, 'Savings', 35000, 'Pune'),
61      (10009, 109, 'Savings', 85000, 'Hyderabad'),
62      (10010, 110, 'Current', 110000, 'Vijayawada');
63 •   select * from Account;

```

Result Grid | Filter Rows:  Edit: Export/Import: Wrap Cell Content

|   | Account_No | Customer_ID | Account_Type | Balance   | Branch     |
|---|------------|-------------|--------------|-----------|------------|
| ▶ | 10001      | 101         | Savings      | 50000.00  | Hyderabad  |
|   | 10002      | 102         | Savings      | 75000.00  | Vijayawada |
|   | 10003      | 103         | Current      | 120000.00 | Bangalore  |
|   | 10004      | 104         | Savings      | 45000.00  | Chennai    |
|   | 10005      | 105         | Current      | 90000.00  | Hyderabad  |
|   | 10006      | 106         | Savings      | 65000.00  | Delhi      |

Account 3 x

Output

Action Output

| #   | Time     | Action                                                                          | Message            |
|-----|----------|---------------------------------------------------------------------------------|--------------------|
| ✖ 1 | 15:36:24 | INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email, City) VALUES... | Error Code: 10     |
| ✔ 2 | 15:37:47 | select * from Customer LIMIT 0, 1000                                            | 10 row(s) returned |
| ✖ 3 | 15:39:55 | INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch...  | Error Code: 10     |
| ✔ 4 | 15:40:19 | select * from Account LIMIT 0, 1000                                             | 10 row(s) returned |

=====

## 8. INSERT TRANSACTION DATA

=====

INSERT INTO Bank\_Transaction

(Account\_No, Transaction\_Type, Amount)

VALUES

(10001, 'DEPOSIT', 10000),

(10001, 'WITHDRAW', 5000),

(10002, 'DEPOSIT', 15000),

(10003, 'WITHDRAW', 20000),

(10004, 'DEPOSIT', 5000),

(10005, 'WITHDRAW', 10000),

(10006, 'DEPOSIT', 12000),

(10007, 'DEPOSIT', 25000),

(10008, 'WITHDRAW', 5000),

(10009, 'DEPOSIT', 20000),

(10010, 'WITHDRAW', 15000);

```
70      (10003, 'WITHDRAW', 20000),
71      (10004, 'DEPOSIT', 5000),
72      (10005, 'WITHDRAW', 10000),
73      (10006, 'DEPOSIT', 12000),
74      (10007, 'DEPOSIT', 25000),
75      (10008, 'WITHDRAW', 5000),
76      (10009, 'DEPOSIT', 20000),
77      (10010, 'WITHDRAW', 15000);
78 • INSERT INTO Loan
79     (Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_Rate)
80     VALUES
81     (501, 101, 'Home Loan', 5000000, 7.5),
82     (502, 102, 'Education Loan', 1000000, 6.5),
```

Output

Action Output

| #   | Time     | Action                                                                          | Message                                                            |
|-----|----------|---------------------------------------------------------------------------------|--------------------------------------------------------------------|
| ❌ 1 | 15:36:24 | INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email, City) VALUES... | Error Code: 1062. Duplicate entry '101' for key 'customer.PRIMARY' |
| ✅ 2 | 15:37:47 | select * from Customer LIMIT 0, 1000                                            | 10 row(s) returned                                                 |
| ❌ 3 | 15:39:55 | INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch...  | Error Code: 1062. Duplicate entry '10001' for key 'account.PRIMAR' |
| ✅ 4 | 15:40:19 | select * from Account LIMIT 0, 1000                                             | 10 row(s) returned                                                 |
| ✅ 5 | 15:41:24 | INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES...   | 11 row(s) affected Records: 11 Duplicates: 0 Warnings: 0           |

=====

## 9. INSERT LOAN DATA

=====

```
INSERT INTO Loan
```

```
(Loan_ID, Customer_ID, Loan_Type,  
Loan_Amount, Interest_Rate)
```

```
VALUES
```

```
(501, 101, 'Home Loan', 5000000, 7.5),
```

```
(502, 102, 'Education Loan', 1000000, 6.5),
```

```
(503, 103, 'Car Loan', 800000, 8.2),
```

```
(504, 104, 'Personal Loan', 500000, 10.5),
```

```
(505, 105, 'Home Loan', 4000000, 7.2),
```

```
(506, 106, 'Car Loan', 900000, 8.5),
```

```
(507, 107, 'Business Loan', 3000000, 9.0),
```

```
(508, 109, 'Personal Loan', 600000, 10.0);
```

```

85  (505, 105, 'Home Loan', 4000000, 7.2),
86  (506, 106, 'Car Loan', 900000, 8.5),
87  (507, 107, 'Business Loan', 3000000, 9.0),
88  (508, 109, 'Personal Loan', 600000, 10.0);
89  SELECT * FROM Loan;

```

| Loan_ID | Customer_ID | Loan_Type      | Loan_Amount | Interest_Rate |
|---------|-------------|----------------|-------------|---------------|
| 501     | 101         | Home Loan      | 5000000.00  | 7.50          |
| 502     | 102         | Education Loan | 1000000.00  | 6.50          |
| 503     | 103         | Car Loan       | 800000.00   | 8.20          |
| 504     | 104         | Personal Loan  | 500000.00   | 10.50         |
| 505     | 105         | Home Loan      | 4000000.00  | 7.20          |
| 506     | 106         | Car Loan       | 900000.00   | 8.50          |

Loan 4 x

Output

| # | Time     | Action                                                                        | Message                                      |
|---|----------|-------------------------------------------------------------------------------|----------------------------------------------|
| 2 | 15:37:47 | select * from Customer LIMIT 0, 1000                                          | 10 row(s) returned                           |
| 3 | 15:39:55 | INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branc... | Error Code: 1062. Duplicate entry '10001     |
| 4 | 15:40:19 | select * from Account LIMIT 0, 1000                                           | 10 row(s) returned                           |
| 5 | 15:41:24 | INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUE...  | 11 row(s) affected Records: 11 Duplicat      |
| 6 | 15:42:44 | INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_R... | Error Code: 1062. Duplicate entry '501' f... |
| 7 | 15:43:07 | SELECT * FROM Loan LIMIT 0, 1000                                              | 8 row(s) returned                            |

## 10. DISPLAY ORIGINAL TABLES

```
SELECT * FROM Customer;
```

```

87      (507, 107, 'Business Loan', 3000000, 9.0),
88      (508, 109, 'Personal Loan', 600000, 10.0);
89 • SELECT * FROM Loan;
90 • SELECT * FROM Customer;

```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

|   | Customer_ID | Customer_Name | Phone      | Email           | City       |
|---|-------------|---------------|------------|-----------------|------------|
| ▶ | 101         | Ravi Kumar    | 9876543210 | ravi@gmail.com  | Hyderabad  |
|   | 102         | Priya Sharma  | 9876543211 | priya@gmail.com | Vijayawada |
|   | 103         | Arjun Reddy   | 9876543212 | arjun@gmail.com | Bangalore  |
|   | 104         | Sneha Rao     | 9876543213 | sneha@gmail.com | Chennai    |
|   | 105         | Kiran Kumar   | 9876543214 | kiran@gmail.com | Hyderabad  |
|   | 106         | Anil Kumar    | 9876543215 | anil@gmail.com  | Delhi      |

Customer 5 x

Output

Action Output

| # | Time       | Action                                                                        | Message                                  |
|---|------------|-------------------------------------------------------------------------------|------------------------------------------|
| ✖ | 3 15:39:55 | INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branc... | Error Code: 1062. Duplicate entry '1000' |
| ✔ | 4 15:40:19 | select * from Account LIMIT 0, 1000                                           | 10 row(s) returned                       |
| ✔ | 5 15:41:24 | INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUE...  | 11 row(s) affected Records: 11 Duplic... |
| ✖ | 6 15:42:44 | INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_R... | Error Code: 1062. Duplicate entry '501'  |
| ✔ | 7 15:43:07 | SELECT * FROM Loan LIMIT 0, 1000                                              | 8 row(s) returned                        |
| ✔ | 8 15:43:33 | SELECT * FROM Customer LIMIT 0, 1000                                          | 10 row(s) returned                       |

SELECT \* FROM Account;

SQL File 4

Limit to 1000 rows

```

87 (507, 107, 'Business Loan', 3000000, 9.0),
88 -- Execute the selected portion of the script or everything, if there is no selection
89 • SELECT * FROM Loan;
90 • SELECT * FROM Customer;
91 • SELECT * FROM Account;

```

Result Grid

|   | Account_No | Customer_ID | Account_Type | Balance   | Branch     |
|---|------------|-------------|--------------|-----------|------------|
| ▶ | 10001      | 101         | Savings      | 50000.00  | Hyderabad  |
|   | 10002      | 102         | Savings      | 75000.00  | Vijayawada |
|   | 10003      | 103         | Current      | 120000.00 | Bangalore  |
|   | 10004      | 104         | Savings      | 45000.00  | Chennai    |
|   | 10005      | 105         | Current      | 90000.00  | Hyderabad  |
|   | 10006      | 106         | Savings      | 65000.00  | Delhi      |

Account 6

Output

Action Output

| #   | Time     | Action                                                                        | Message                           |
|-----|----------|-------------------------------------------------------------------------------|-----------------------------------|
| ✓ 4 | 15:40:19 | select * from Account LIMIT 0, 1000                                           | 10 row(s) returned                |
| ✓ 5 | 15:41:24 | INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUE...  | 11 row(s) affected Records: 11    |
| ✗ 6 | 15:42:44 | INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_R... | Error Code: 1062. Duplicate entry |
| ✓ 7 | 15:43:07 | SELECT * FROM Loan LIMIT 0, 1000                                              | 8 row(s) returned                 |
| ✓ 8 | 15:43:33 | SELECT * FROM Customer LIMIT 0, 1000                                          | 10 row(s) returned                |
| ✓ 9 | 15:43:51 | SELECT * FROM Account LIMIT 0, 1000                                           | 10 row(s) returned                |

SELECT \* FROM Bank\_Transaction;

```

89 • SELECT * FROM Loan;
90 • SELECT * FROM Customer;
91 • SELECT * FROM Account;
92 • SELECT * FROM Bank_Transaction;
93 • SELECT * FROM Loan;

```

Result Grid

|   | Transaction_ID | Account_No | Transaction_Type | Amount   | Transaction_Date    |
|---|----------------|------------|------------------|----------|---------------------|
| ▶ | 23             | 10001      | DEPOSIT          | 10000.00 | 2026-08-12 15:22:14 |
|   | 24             | 10001      | WITHDRAW         | 5000.00  | 2026-08-12 15:22:14 |
|   | 25             | 10002      | DEPOSIT          | 15000.00 | 2026-08-12 15:22:14 |
|   | 26             | 10003      | WITHDRAW         | 20000.00 | 2026-08-12 15:22:14 |
|   | 27             | 10004      | DEPOSIT          | 5000.00  | 2026-08-12 15:22:14 |
|   | 28             | 10005      | WITHDRAW         | 10000.00 | 2026-08-12 15:22:14 |

Bank\_Transaction 7

Output

Action Output

| #    | Time     | Action                                                                        | Message                                                        |
|------|----------|-------------------------------------------------------------------------------|----------------------------------------------------------------|
| ✓ 5  | 15:41:24 | INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUE...  | 11 row(s) affected Records: 11 Duplicates: 0 Warnings: 0       |
| ✗ 6  | 15:42:44 | INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_R... | Error Code: 1062. Duplicate entry '501' for key 'loan.PRIMARY' |
| ✓ 7  | 15:43:07 | SELECT * FROM Loan LIMIT 0, 1000                                              | 8 row(s) returned                                              |
| ✓ 8  | 15:43:33 | SELECT * FROM Customer LIMIT 0, 1000                                          | 10 row(s) returned                                             |
| ✓ 9  | 15:43:51 | SELECT * FROM Account LIMIT 0, 1000                                           | 10 row(s) returned                                             |
| ✓ 10 | 15:44:13 | SELECT * FROM Bank_Transaction LIMIT 0, 1000                                  | 22 row(s) returned                                             |

SELECT \* FROM Loan;

The screenshot shows a database management interface with a SQL editor at the top containing five queries. Below the editor is a 'Result Grid' showing the results of the first query, 'SELECT \* FROM Loan;'. The grid has columns for Loan\_ID, Customer\_ID, Loan\_Type, Loan\_Amount, and Interest\_Rate. The data is as follows:

| Loan_ID | Customer_ID | Loan_Type      | Loan_Amount | Interest_Rate |
|---------|-------------|----------------|-------------|---------------|
| 501     | 101         | Home Loan      | 5000000.00  | 7.50          |
| 502     | 102         | Education Loan | 800000.00   | 6.50          |
| 503     | 103         | Car Loan       | 800000.00   | 8.20          |
| 504     | 104         | Personal Loan  | 500000.00   | 10.50         |
| 505     | 105         | Home Loan      | 4000000.00  | 7.20          |
| 506     | 106         | Car Loan       | 900000.00   | 8.50          |

Below the result grid is an 'Output' section with a tab labeled 'Action Output'. It shows a log of database actions:

| #  | Time     | Action                                                                        | Message                                       |
|----|----------|-------------------------------------------------------------------------------|-----------------------------------------------|
| 6  | 15:42:44 | INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_R... | Error Code: 1062. Duplicate entry '501' for l |
| 7  | 15:43:07 | SELECT * FROM Loan LIMIT 0, 1000                                              | 8 row(s) returned                             |
| 8  | 15:43:33 | SELECT * FROM Customer LIMIT 0, 1000                                          | 10 row(s) returned                            |
| 9  | 15:43:51 | SELECT * FROM Account LIMIT 0, 1000                                           | 10 row(s) returned                            |
| 10 | 15:44:13 | SELECT * FROM Bank_Transaction LIMIT 0, 1000                                  | 22 row(s) returned                            |
| 11 | 15:44:33 | SELECT * FROM Loan LIMIT 0, 1000                                              | 8 row(s) returned                             |

=====

## PART A – SIMPLE VIEWS

=====

-----

### 11. CREATE A VIEW FOR ALL CUSTOMERS

-----

CREATE VIEW Customer\_View AS

SELECT \*

FROM Customer;

DISPLAY VIEW:

SELECT \* FROM Customer\_View;

94 • CREATE VIEW Customer\_Basic\_View AS

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

|   | Customer_ID | Customer_Name | City        |
|---|-------------|---------------|-------------|
| ▶ | 101         | Ravi Kumar    | Hyderabad   |
|   | 102         | Priya Sharma  | Vijayawada  |
|   | 103         | Arjun Reddy   | Bangalore   |
|   | 104         | Sneha Rao     | Chennai     |
|   | 105         | Kiran Kumar   | Kiran Kumar |
|   | 106         | Anil Kumar    | Dehi        |

Customer\_Basic\_View 9 x

Output

Action Output

| #    | Time     | Action                                                                    | Message            |
|------|----------|---------------------------------------------------------------------------|--------------------|
| ✓ 8  | 15:43:33 | SELECT * FROM Customer LIMIT 0, 1000                                      | 10 row(s) returned |
| ✓ 9  | 15:43:51 | SELECT * FROM Account LIMIT 0, 1000                                       | 10 row(s) returned |
| ✓ 10 | 15:44:13 | SELECT * FROM Bank_Transaction LIMIT 0, 1000                              | 22 row(s) returned |
| ✓ 11 | 15:44:33 | SELECT * FROM Loan LIMIT 0, 1000                                          | 8 row(s) returned  |
| ✓ 12 | 15:45:20 | CREATE VIEW Customer_Basic_View AS SELECT Customer_ID, Customer_Name, ... | 0 row(s) affected  |
| ✓ 13 | 15:45:20 | SELECT * FROM Customer_Basic_View LIMIT 0, 1000                           | 10 row(s) returned |

-----

## 12. CREATE A VIEW FOR CUSTOMER BASIC INFORMATION

-----

CREATE VIEW Customer\_Basic\_View AS

SELECT

Customer\_ID,

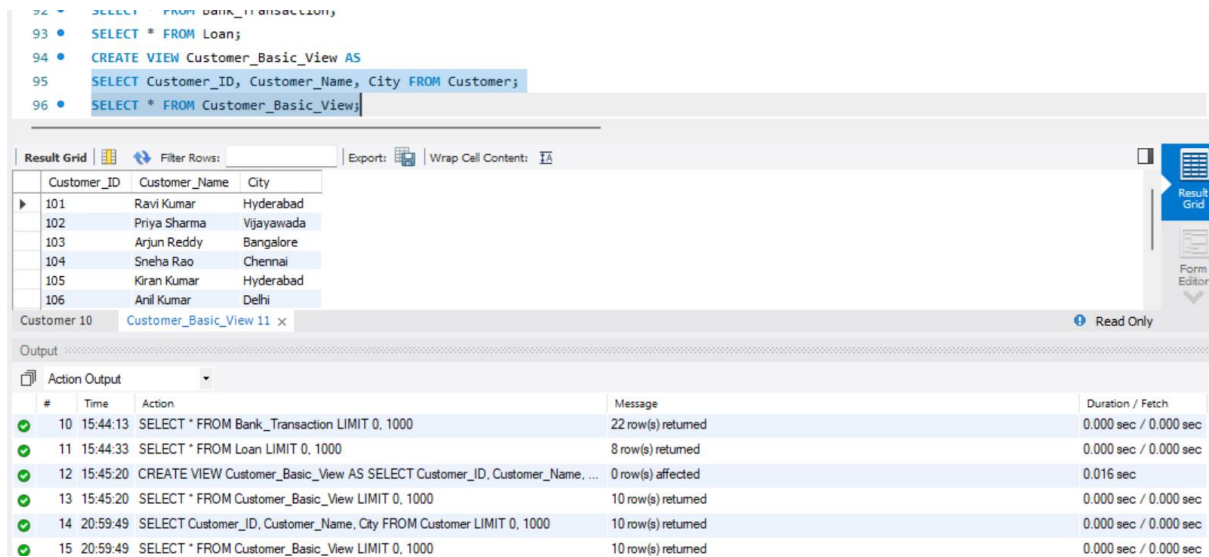
Customer\_Name,

City

FROM Customer;

DISPLAY:

```
SELECT * FROM Customer_Basic_View;
```



SQL Editor:

```
92 SELECT * FROM Bank_Transaction;  
93 SELECT * FROM Loan;  
94 CREATE VIEW Customer_Basic_View AS  
95 SELECT Customer_ID, Customer_Name, City FROM Customer;  
96 SELECT * FROM Customer_Basic_View;
```

Result Grid:

| Customer_ID | Customer_Name | City       |
|-------------|---------------|------------|
| 101         | Ravi Kumar    | Hyderabad  |
| 102         | Priya Sharma  | Vijayawada |
| 103         | Arjun Reddy   | Bangalore  |
| 104         | Sneha Rao     | Chennai    |
| 105         | Kiran Kumar   | Hyderabad  |
| 106         | Anil Kumar    | Delhi      |

Output:

| #  | Time     | Action                                                                    | Message            | Duration / Fetch      |
|----|----------|---------------------------------------------------------------------------|--------------------|-----------------------|
| 10 | 15:44:13 | SELECT * FROM Bank_Transaction LIMIT 0, 1000                              | 22 row(s) returned | 0.000 sec / 0.000 sec |
| 11 | 15:44:33 | SELECT * FROM Loan LIMIT 0, 1000                                          | 8 row(s) returned  | 0.000 sec / 0.000 sec |
| 12 | 15:45:20 | CREATE VIEW Customer_Basic_View AS SELECT Customer_ID, Customer_Name, ... | 0 row(s) affected  | 0.016 sec             |
| 13 | 15:45:20 | SELECT * FROM Customer_Basic_View LIMIT 0, 1000                           | 10 row(s) returned | 0.000 sec / 0.000 sec |
| 14 | 20:59:49 | SELECT Customer_ID, Customer_Name, City FROM Customer LIMIT 0, 1000       | 10 row(s) returned | 0.000 sec / 0.000 sec |
| 15 | 20:59:49 | SELECT * FROM Customer_Basic_View LIMIT 0, 1000                           | 10 row(s) returned | 0.000 sec / 0.000 sec |

### 13. CREATE A VIEW FOR ACCOUNT INFORMATION

```
CREATE VIEW Account_View AS
```

```
SELECT
```

```
Account_No,
```

```
Account_Type,
```

```
Balance,
```

```
Branch
```

```
FROM Account;
```

DISPLAY:

```
SELECT * FROM Account_View;
```

```

101     Account_Type,
102     Balance,
103     Branch
104 FROM Account;
105 SELECT * FROM Account_View;

```

Result Grid | Filter Rows:  | Export: | Wrap Cell Content:

|   | Account_No | Account_Type | Balance   | Branch     |
|---|------------|--------------|-----------|------------|
| ▶ | 10001      | Savings      | 50000.00  | Hyderabad  |
|   | 10002      | Savings      | 75000.00  | Vijayawada |
|   | 10003      | Current      | 120000.00 | Bangalore  |
|   | 10004      | Savings      | 45000.00  | Chennai    |
|   | 10005      | Current      | 90000.00  | Hyderabad  |
|   | 10006      | Savings      | 65000.00  | Delhi      |

Account\_View 12 x

Output

Action Output

| #    | Time     | Action                                                              | Message                                   |
|------|----------|---------------------------------------------------------------------|-------------------------------------------|
| ✓ 13 | 15:45:20 | SELECT * FROM Customer_Basic_View LIMIT 0, 1000                     | 10 row(s) returned                        |
| ✓ 14 | 20:59:49 | SELECT Customer_ID, Customer_Name, City FROM Customer LIMIT 0, 1000 | 10 row(s) returned                        |
| ✓ 15 | 20:59:49 | SELECT * FROM Customer_Basic_View LIMIT 0, 1000                     | 10 row(s) returned                        |
| ✗ 16 | 22:33:43 | FROM Account                                                        | Error Code: 1064. You have an error in yo |
| ✓ 17 | 22:34:35 | CREATE VIEW Account_View AS SELECT Account_No, Account_Type, B...   | 0 row(s) affected                         |
| ✓ 18 | 22:34:35 | SELECT * FROM Account_View LIMIT 0, 1000                            | 10 row(s) returned                        |

#### 14. CREATE A VIEW FOR SAVINGS ACCOUNTS

CREATE VIEW Savings\_Account\_View AS

SELECT \*

FROM Account

WHERE Account\_Type = 'Savings';

DISPLAY:

SELECT \* FROM Savings\_Account\_View;

SQL Editor

```

107 CREATE VIEW Savings_Account_View AS
108 SELECT *
109 FROM Account
110 WHERE Account_Type = 'Savings';
111 SELECT * FROM Savings_Account_View;

```

Result Grid

| Account_No | Customer_ID | Account_Type | Balance  | Branch     |
|------------|-------------|--------------|----------|------------|
| 10001      | 101         | Savings      | 50000.00 | Hyderabad  |
| 10002      | 102         | Savings      | 75000.00 | Vijayawada |
| 10004      | 104         | Savings      | 45000.00 | Chennai    |
| 10006      | 106         | Savings      | 65000.00 | Delhi      |
| 10008      | 108         | Savings      | 35000.00 | Pune       |
| 10009      | 109         | Savings      | 85000.00 | Hyderabad  |

Savings\_Account\_View 13 x

Output

| #  | Time     | Action                                                                | Message                                                                               | Durati |
|----|----------|-----------------------------------------------------------------------|---------------------------------------------------------------------------------------|--------|
| 15 | 20:59:49 | SELECT * FROM Customer_Basic_View LIMIT 0, 1000                       | 10 row(s) returned                                                                    | 0.000  |
| 16 | 22:33:43 | FROM Account                                                          | Error Code: 1064. You have an error in your SQL syntax; check the manual that corr... | 0.000  |
| 17 | 22:34:35 | CREATE VIEW Account_View AS SELECT Account_No, Account_Type, B...     | 0 row(s) affected                                                                     | 0.016  |
| 18 | 22:34:35 | SELECT * FROM Account_View LIMIT 0, 1000                              | 10 row(s) returned                                                                    | 0.000  |
| 19 | 22:35:47 | CREATE VIEW Savings_Account_View AS SELECT * FROM Account WHERE Ac... | 0 row(s) affected                                                                     | 0.016  |
| 20 | 22:35:47 | SELECT * FROM Savings_Account_View LIMIT 0, 1000                      | 6 row(s) returned                                                                     | 0.000  |

## 15. CREATE A VIEW FOR CURRENT ACCOUNTS

CREATE VIEW Current\_Account\_View AS

SELECT \*

FROM Account

WHERE Account\_Type = 'Current';

DISPLAY:

SELECT \* FROM Current\_Account\_View;

```

114 SELECT *
115 FROM Account
116 WHERE Account_Type = 'Current';
117 • SELECT * FROM Current_Account_View;
118

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

|   | Account_No | Customer_ID | Account_Type | Balance   | Branch     |
|---|------------|-------------|--------------|-----------|------------|
| ▶ | 10003      | 103         | Current      | 120000.00 | Bangalore  |
|   | 10005      | 105         | Current      | 90000.00  | Hyderabad  |
|   | 10007      | 107         | Current      | 150000.00 | Mumbai     |
|   | 10010      | 110         | Current      | 110000.00 | Vijayawada |

Current\_Account\_View 14 x

Output

Action Output

| #    | Time     | Action                                                                 | Message            |
|------|----------|------------------------------------------------------------------------|--------------------|
| ✓ 17 | 22:34:35 | CREATE VIEW Account_View AS SELECT Account_No, Account_Type, B...      | 0 row(s) affected  |
| ✓ 18 | 22:34:35 | SELECT * FROM Account_View LIMIT 0, 1000                               | 10 row(s) returned |
| ✓ 19 | 22:35:47 | CREATE VIEW Savings_Account_View AS SELECT * FROM Account WHERE Ac...  | 0 row(s) affected  |
| ✓ 20 | 22:35:47 | SELECT * FROM Savings_Account_View LIMIT 0, 1000                       | 6 row(s) returned  |
| ✓ 21 | 22:36:40 | CREATE VIEW Current_Account_View AS SELECT * FROM Account WHERE Acc... | 0 row(s) affected  |
| ✓ 22 | 22:36:40 | SELECT * FROM Current_Account_View LIMIT 0, 1000                       | 4 row(s) returned  |

## PART B – VIEWS WITH CONDITIONS

### 16. HIGH BALANCE ACCOUNT VIEW

CREATE VIEW High\_Balance\_View AS

SELECT

Account\_No,

Customer\_ID,

Account\_Type,

Balance

FROM Account

WHERE Balance > 100000;

DISPLAY:

SELECT \* FROM High\_Balance\_View;

The screenshot displays a database management interface. At the top, a SQL query is entered in a text area:

```
120 Balance
121 FROM Account
122 WHERE Balance > 100000;
123 • SELECT * FROM High_Balance_View;
124
```

Below the query editor, the 'Result Grid' shows the output of the query. It contains a table with the following data:

| Account_No | Customer_ID | Account_Type | Balance   |
|------------|-------------|--------------|-----------|
| 10003      | 103         | Current      | 120000.00 |
| 10007      | 107         | Current      | 150000.00 |
| 10010      | 110         | Current      | 110000.00 |

Below the result grid, the 'Output' pane shows the 'Action Output' log. It contains the following entries:

| #  | Time     | Action                                                                 | Message           |
|----|----------|------------------------------------------------------------------------|-------------------|
| 19 | 22:35:47 | CREATE VIEW Savings_Account_View AS SELECT * FROM Account WHERE Ac...  | 0 row(s) affected |
| 20 | 22:35:47 | SELECT * FROM Savings_Account_View LIMIT 0, 1000                       | 6 row(s) returned |
| 21 | 22:36:40 | CREATE VIEW Current_Account_View AS SELECT * FROM Account WHERE Acc... | 0 row(s) affected |
| 22 | 22:36:40 | SELECT * FROM Current_Account_View LIMIT 0, 1000                       | 4 row(s) returned |
| 23 | 22:37:29 | CREATE VIEW High_Balance_View AS SELECT Account_No, Customer_ID, ...   | 0 row(s) affected |
| 24 | 22:37:29 | SELECT * FROM High_Balance_View LIMIT 0, 1000                          | 3 row(s) returned |

## 17. HYDERABAD ACCOUNT VIEW

CREATE VIEW Hyderabad\_Account\_View AS

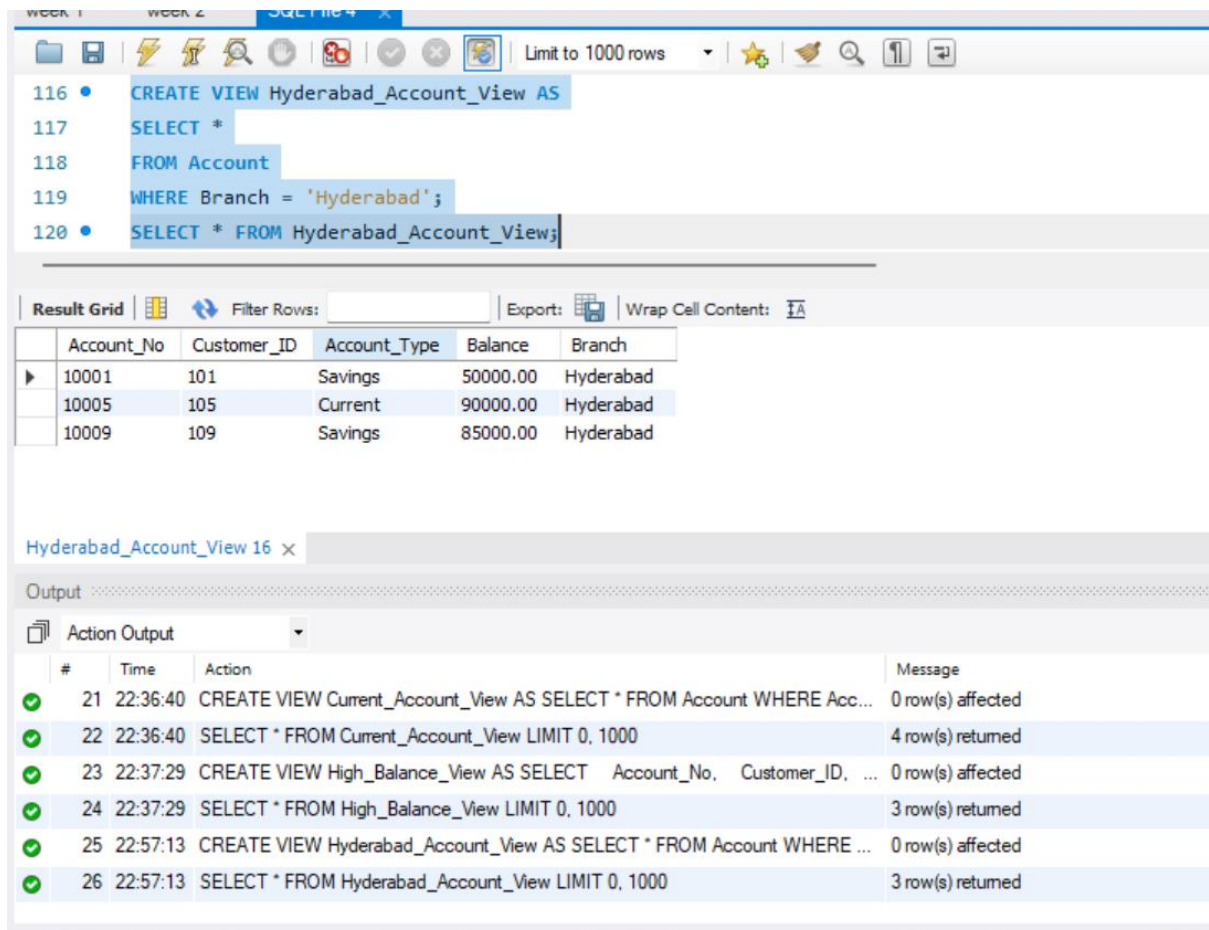
SELECT \*

FROM Account

WHERE Branch = 'Hyderabad';

DISPLAY:

```
SELECT * FROM Hyderabad_Account_View;
```



The screenshot shows a SQL IDE interface with the following components:

- SQL Editor:** Contains the following SQL code:

```
116 • CREATE VIEW Hyderabad_Account_View AS
117 SELECT *
118 FROM Account
119 WHERE Branch = 'Hyderabad';
120 • SELECT * FROM Hyderabad_Account_View;
```
- Results Grid:** Displays the result of the SELECT query. It has columns: Account\_No, Customer\_ID, Account\_Type, Balance, and Branch. The data is as follows:

| Account_No | Customer_ID | Account_Type | Balance  | Branch    |
|------------|-------------|--------------|----------|-----------|
| 10001      | 101         | Savings      | 50000.00 | Hyderabad |
| 10005      | 105         | Current      | 90000.00 | Hyderabad |
| 10009      | 109         | Savings      | 85000.00 | Hyderabad |
- Output Pane:** Shows the execution log with the following entries:

| #  | Time     | Action                                                                 | Message           |
|----|----------|------------------------------------------------------------------------|-------------------|
| 21 | 22:36:40 | CREATE VIEW Current_Account_View AS SELECT * FROM Account WHERE Acc... | 0 row(s) affected |
| 22 | 22:36:40 | SELECT * FROM Current_Account_View LIMIT 0, 1000                       | 4 row(s) returned |
| 23 | 22:37:29 | CREATE VIEW High_Balance_View AS SELECT Account_No, Customer_ID, ...   | 0 row(s) affected |
| 24 | 22:37:29 | SELECT * FROM High_Balance_View LIMIT 0, 1000                          | 3 row(s) returned |
| 25 | 22:57:13 | CREATE VIEW Hyderabad_Account_View AS SELECT * FROM Account WHERE ...  | 0 row(s) affected |
| 26 | 22:57:13 | SELECT * FROM Hyderabad_Account_View LIMIT 0, 1000                     | 3 row(s) returned |

## 18. LOW BALANCE ACCOUNT VIEW

```
CREATE VIEW Low_Balance_View AS
```

```
SELECT
```

```
Account_No,
```

```
Customer_ID,
```

```
Balance
```

```
FROM Account
```

WHERE Balance < 50000;

DISPLAY:

SELECT \* FROM Low\_Balance\_View;

CREATE VIEW Low\_Balance\_View AS

SELECT

Account\_No,

Customer\_ID,

Balance

FROM Account

WHERE Balance < 50000;

DISPLAY:

SELECT \* FROM Low\_Balance\_View;

=====

## PART C – VIEWS USING JOINS

=====

-----

### 19. CUSTOMER + ACCOUNT VIEW

-----

```
CREATE VIEW Customer_Account_View AS
```

```
SELECT
```

```
    C.Customer_ID,
```

```
    C.Customer_Name,
```

```
    C.City,
```

```
    A.Account_No,
```

```
    A.Account_Type,
```

```
    A.Balance,
```

```
    A.Branch
```

```
FROM Customer C
```

```
JOIN Account A
```

```
ON C.Customer_ID = A.Customer_ID;
```

DISPLAY:

```
SELECT * FROM Customer_Account_View;
```

The screenshot shows a SQL IDE interface with a query editor at the top and a result grid below. The query editor contains the following SQL code:

```
120 A.Branch
121 FROM Customer C
122 JOIN Account A
123 ON C.Customer_ID = A.Customer_ID;
124 SELECT * FROM Customer_Account_View;
```

The result grid displays the following data:

| Customer_ID | Customer_Name | City       | Account_No | Account_Type | Balance   | Branch     |
|-------------|---------------|------------|------------|--------------|-----------|------------|
| 101         | Ravi Kumar    | Hyderabad  | 10001      | Savings      | 50000.00  | Hyderabad  |
| 102         | Priya Sharma  | Vijayawada | 10002      | Savings      | 75000.00  | Vijayawada |
| 103         | Arjun Reddy   | Bangalore  | 10003      | Current      | 120000.00 | Bangalore  |
| 104         | Sneha Rao     | Chennai    | 10004      | Savings      | 45000.00  | Chennai    |
| 105         | Kiran Kumar   | Hyderabad  | 10005      | Current      | 90000.00  | Hyderabad  |
| 106         | Anil Kumar    | Delhi      | 10006      | Savings      | 65000.00  | Delhi      |

Below the result grid, the 'Output' section shows the execution log:

| #  | Time     | Action                                                                | Message            |
|----|----------|-----------------------------------------------------------------------|--------------------|
| 25 | 22:57:13 | CREATE VIEW Hyderabad_Account_View AS SELECT * FROM Account WHERE ... | 0 row(s) affected  |
| 26 | 22:57:13 | SELECT * FROM Hyderabad_Account_View LIMIT 0, 1000                    | 3 row(s) returned  |
| 27 | 22:58:05 | CREATE VIEW Low_Balance_View AS SELECT Account_No, Customer_ID, ...   | 0 row(s) affected  |
| 28 | 22:58:05 | SELECT * FROM Low_Balance_View LIMIT 0, 1000                          | 2 row(s) returned  |
| 29 | 22:58:43 | CREATE VIEW Customer_Account_View AS SELECT C.Customer_ID, C.Custo... | 0 row(s) affected  |
| 30 | 22:58:43 | SELECT * FROM Customer_Account_View LIMIT 0, 1000                     | 10 row(s) returned |

## 20. HYDERABAD CUSTOMER ACCOUNT VIEW

---

CREATE VIEW Hyderabad\_Customer\_Accounts AS

SELECT

C.Customer\_Name,

C.City,

A.Account\_No,

A.Account\_Type,

A.Balance

FROM Customer C

JOIN Account A

ON C.Customer\_ID = A.Customer\_ID

WHERE A.Branch = 'Hyderabad';

DISPLAY:

SELECT \* FROM Hyderabad\_Customer\_Accounts;

